



**RAMAKRISHNA
MATH
MEDIA LAB
@CHENNAI
ideas for innovation**

Team: SatyaBharat

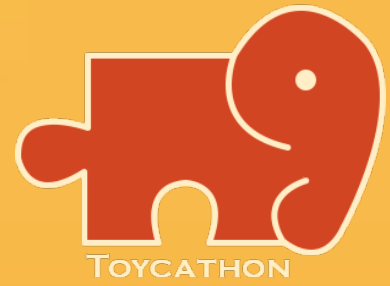


Narendra

A Smart Toy to analyze the Emotional well-being of children through Facial analysis and empowering them with Mind fitness training such as Meditation game and narration of Inspirational stories

Last Updated: May 22nd, 2022

Any Questions? Email us at medialab@chennaiamath.org



*Narendra
Smart Toy*

Protecting Humanity's Greatest Assets

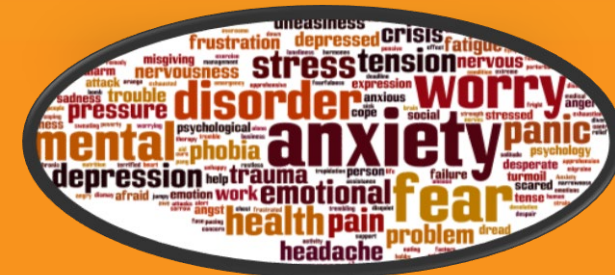
Q: Narendra, what are the greatest assets of Humanity and how do we protect them?

A: *Our Primary asset in the world is children. If they are healthy, everyone is happy. Today, all over the world, there is a greater need to protect the mental health of children. I see that many of the modern psychological & emotional problems that humans face, can be solved with an ancient approach to education. With our ancient Indian system, we could teach children the Life-Skills that enable them to have the mental strength to boldly face the difficulties of the world. Over time, they would master the “**Art of becoming and being a human**”.*



Indian Education System & Effects of Covid-19 lockdown

- India has one of the world's largest education systems in the world with nearly 300 million students (Nanda, 2020) and Covid-19 lockdown has significantly impacted millions of children from both rural and urban areas (Singh et al., 2020)
- There are also 40 million children coming from marginalized families affected due to poverty, social inequality, education and digital divide
- Recently, a growing number of concerns related to emotional health and issues of these children such as anxiety, acute stress, loneliness, have accelerated due to lockdown



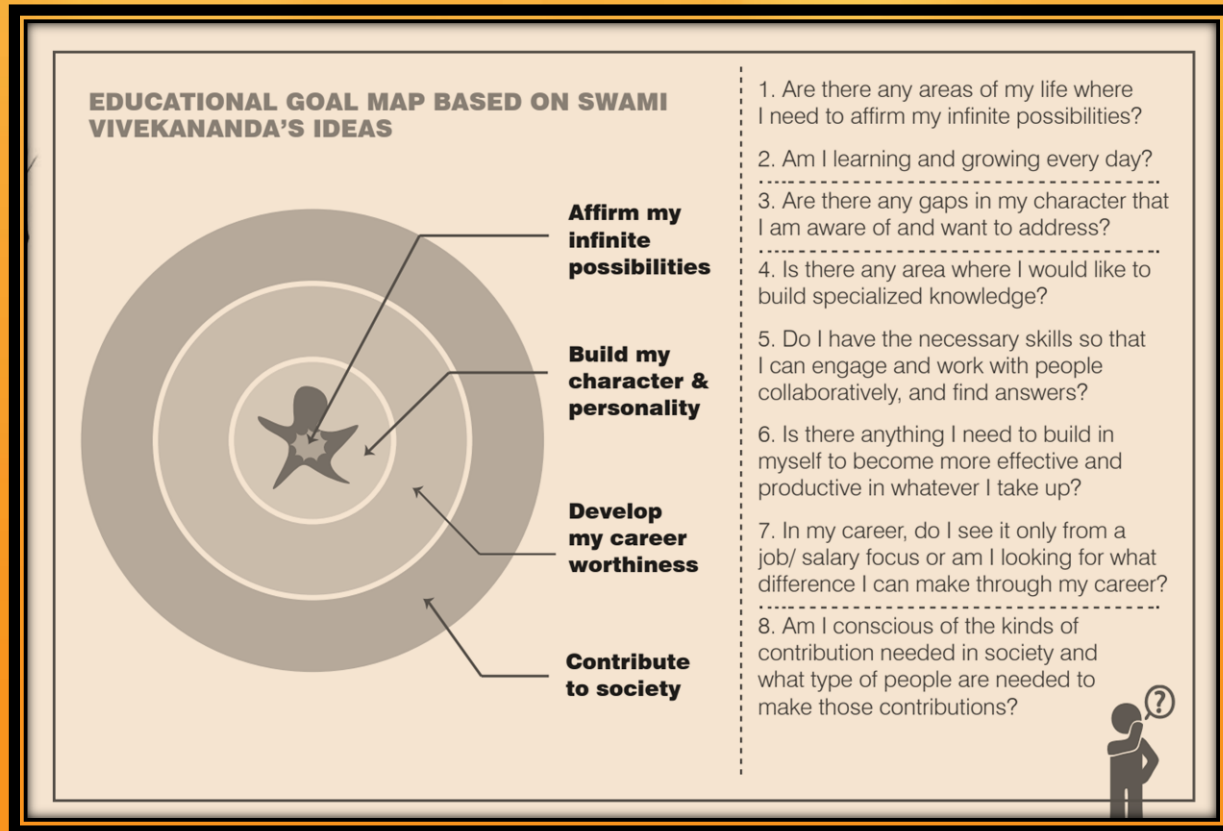
Identification of Emotional Health of Children

- Multi-modal solutions like real-time Facial Analysis among children (Simcock et al., 2020) and voice tonal analysis helps in analyzing the emotional well-being of children



Protecting the emotional Well-being of Children (Life Skills)

- A Life Skills program (Mahamedhananda & Srinivas, 2018) building one's character and personality with guidance/inspiration from positive Real-life Role models influence the behavioral psychology of children (Greenberger, Chen, & Beam, 1998)



A Role Model...

Swami Vivekananda is the greatest example & role model of one who has manifested his infinite possibilities

“ In the two words, *equilibrium and synthesis*, Vivekananda's constructive genius may be summed up. He embraced all the paths of the spirit: the four yogas in their entirety, renunciation and service, art and science, religion and action, from the most spiritual to the most practical. Each of the ways that he taught had its own limits, but he himself had been through them all, and embraced them all. As in a quadriga, he held the reins of all four ways of truth, and travelled towards Unity along them all simultaneously. He was the personification of the harmony of all human energy. ”

Excerpt from “Life of Vivekananda” published in 1947, by Romain Rolland, the 1915 Nobel Laureate in Literature.

Physical possibilities

From his childhood, Swami Vivekananda excelled in a wide range of games and physical activities. This gave him the capacity to go through enormous physical hardships such as going without food, living in all kinds of surroundings and facing all kinds of weathers.

Creative possibilities

Swami Vivekananda had a wonderful gift of expression and language. He was a musician, singer, composer, and a designer. He had a keen sense of art. He was also an excellent cook. Whatever he put his mind to, he excelled.

Leadership possibilities

Swami Vivekananda was a leader who awakened people. He envisioned a new India, and inspired and led hundreds of people, including many future leaders of the country, towards this vision. He also organized & set up a new spiritual order which now functions as the Ramakrishna Math and Ramakrishna Mission.

Intellectual possibilities

Swami Vivekananda's breadth and depth of knowledge was extraordinary. His deep perception allowed him to go into the very essence of everything. He had the ability to

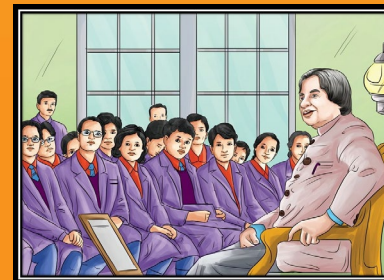
Protecting the emotional Well-being of Children (IBMT)

- Integrated Body Mind Training program utilizing traditional training methods like Yoga, Meditation (Tang, Jiang, & Tang, 2017) & Playtime protect the emotional well-being of children (Johnson et al., 2020)



Need for a Positive Role Model for Children

- Adolescents look up to role models while forming their identities and Positive role models have made positive outcomes in adolescent behavior (Hurd & Zimmerman, 2011)
- Adolescents whose only role model is an entertainer (e.g., singer, actor, etc.) are at higher risk for engaging in smoking, drug use, and fighting later in their lives (Wise, 2019)
- Children expects positive role models age groups slightly older than them yet have the understanding and guidance which a friend would provide (Gozali-Lee, 1994)
- A Role model needs to transcend beyond gender, culture, social status, religion and nation so that it would be all inclusive and would appeal to all children
- Modern kids need a role model who can be a bridge between Traditional & Modern cultures
- Role Model would emphasize growth in overall development (physical, intellectual and emotional) and enable children to have the mental strength and boldly face the difficulties of the world



Need for Socio-Cultural Toys for Kids emotional well-being

- Play provides a healing power and improves the emotional well-being of children (Singer, 1994) and toys play a very important role in facilitating this one (Johnson et al., 2020)
- Socio-Culture deprivation among children results in low self-esteem and low emotional maturity (Jiloha, 2010)
- Social Toys help express and understand emotions in children, also help them build social skills, aids in overall emotional and social development (Seema & Ravi, 2020)
- Research Studies reveal that if children deprived of play suffer emotionally both in present and long term (Guha & Murthy, 2020)

**informatics**



Article

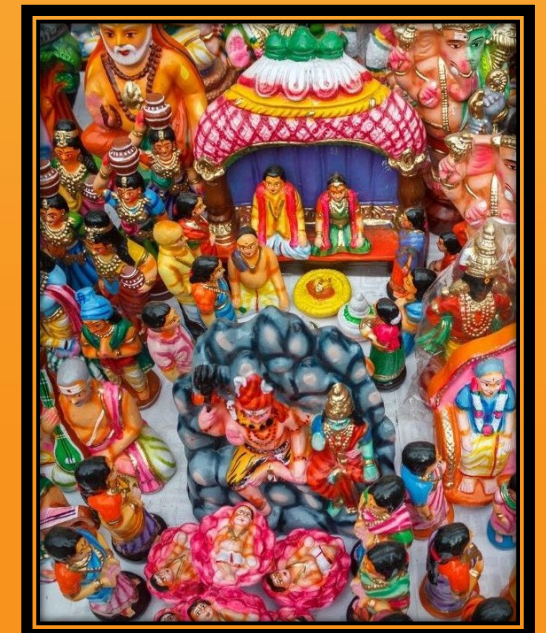
An Affective and Cognitive Toy to Support Mood Disorders

Esperanza Johnson ^{1,*}, Iván González ¹, Tania Mondéjar ^{1,2}, Luis Cabañero-Gómez ¹,
Jesús Fontecha ¹ and Ramón Hervás ¹

¹ Department of Technologies and Information Systems, University of Castilla-La Mancha, 13071 Ciudad Real, Spain; Ivan.GDiaz@uclm.es (I.G.); Tania.Mondejar@esmile.es (T.M.); Profesor.LCGomez@uclm.es (L.C.-G.); Jesus.Fontecha@uclm.es (J.F.); Ramon.HLucas@uclm.es (R.H.)

² eSmile, Psychology for Children & Adolescents, 13071 Ciudad Real, Spain

* Correspondence: MEsperanza.Johnson@uclm.es; Tel.: +34-92-629-5300



Narendra – An Ideal Role Model

- Swami Vivekananda (Childhood Name – Naren) is an ideal role model for young generation in India and his birth anniversary is celebrated as National Youth Day across the country
- Naren - Avid reader, active in games and sports, powerful speaker and good story teller, Great passion for music and an Excellent cook (Chakraborty, 2009)
- Sweet and charming, friendly with everyone, leadership qualities through self-sacrifice, bravery and mental strength, unselfish and had spirit of service
- Attraction to Indian cultural influences such as Ramayana and Mahabharatha
- Reverence for women and equal respect for people across caste, religion & nationalities
- Expert Meditator who regulated his emotions that enabled him to understand the complexities of Life
- Scientific and rational – in later years - had discussions with Nikola Tesla related to Energy & Matter, influenced Jamsetji Tata to setup the Indian Institute of Science
- Truth seeker & Disdain for rumors – always sought the truth & never accepted anything without testing and experiencing for himself

Narendra – An Ideal Role Model



Naren in deep meditation and focus that enabled him to possess extraordinary mental strength and powers to understand complex subject of Life



Naren applies logical reasoning and exhibits fearlessness, he doesn't believe in Grandfather's story of ghost residing in the tree to keep Naren & friends from playing in the garden

Narendra – An Ideal Role Model



*Naren – helping those in need
exhibiting empathetic behavior*



*Naren – a great storyteller
- Learning Ramayana and
Mahabharata at an early age*



*Naren with his self-sacrifice rescuing a
child who is about to get hurt by a
horse carriage*

What is Narendra Smart Toy ?

A Smart Toy to analyze the Emotional well-being of children through Facial analysis and empowering them with Mind fitness training such as Meditation game and narration of Inspirational stories



Narendra Smart Toy – Identifying Emotional Well-being

- A fiber doll of Naren has been designed and a camera, LCD display and mic is connected thereby making a Narendra prototype
- An integrated AI based Camera sensor with LCD display connected to Latte Panda running localized multi-modal emotion detection using facial analysis is prototyped
- Open Source software implemented in Python such as Deep Face (Serengil & Ozpinar, 2020) is utilized for facial analysis
- Once the emotion detection values stream from the system, a negative emotion index is computed real-time and averaged over a specified time period
- If emotion index is less than a specified threshold, specific curated content (Example: Inspirational stories) is played out to the child to lift the child's spirits
- Meditation game is also played out where the child has to sit in a proper posture closing it's eyes for a specified duration

Narendra Smart Toy – Aspects & Apps

- Three human aspects (Thought, Words, Action) are taken - Based on the ideas derived from Integrated body-mind training and Life Skills training, the following are incorporated

Human Aspect	Moral Aspect	Aspects incorporated in Smart Toy	App (Curated Content)
Thought	Right Thinking	Positive Thoughts/Concentration/Meditation	Meditation App
Words	Right Speech	Inspirational and Cultural stories	Story Telling App (Story of Two frogs)
Action	Right Deeds	Role Play situations in Video	Role Play App (Mind your Mind Video) – in Development

Watch your thoughts;
they become words.
Watch your words;
they become actions.
Watch your actions;
they become habits.
Watch your habits;
they become character.
Watch your character;
for it becomes your destiny.



What
would
you
do?

Narendra Implementation – Meditation

- A meditation app helps the child to practice concentration of thought
- The child starts the meditation session by looking at Narendra who is standing in front of him
- The smart toy system observes the initial eye points of child through camera sensor and If the child gets distracted either by swaying to and fro (or) opening his eye, the system triggers a motion alert
- With the motion alert, Narendra gently reminds the child to sit still and tries to motivate him to sit in meditation
- If the child sits still for a specified period, reward points are given to encourage the child



EYE TRACKING

Narendra Implementation – Story Telling

- The StoryTelling App helps the child regulate the emotions with inspiring stories
- The child starts a cultural story session by keeping Naren Bhaiya to his side asking for an inspirational story
- It then displays the relevant video of the inspirational story



Narendra Smart Toy – its inherent nature and uniqueness

How is our toy different from others?

Toys have typically been a source of learning (or) entertainment. But we feel a toy, although inanimate in a certain sense, need to be next to the role of a closer friend (or) sibling connecting to children in an emotional manner.

Rather than seeing toys as an accessory, our toy Narendra would be more of a primary necessity taking care of the emotional well-being of a child.

In short, we see such **TOYS** as **Transformation Of YourSELF**

Does our Toy Reflect Indian Culture, Ethos, Diversity, Psychology?

Yes, Yes, Yes and Yes in a big way. Narendra was a bridge between ancient India culture and modern culture. He has special attraction for Indian cultural influences such as Ramayana and Mahabharatha. Also, great Reverence for women and equal respect for people across caste, religion & nationalities encompasses the diversity. Narendra's plan for education has always been about life-skills development and character building which reflects the ethos. Narendra has always been the one who want our children to be strong from their childhood. He wants them to be bold and wanted them to stand on their own feet independently. So, we strongly believe a TOY with Narendra as Role Model will be a fulfillment of “Arise Awake” and “Be Fearless” for the young generation.

Team Profile

Team Objective

To enable the spirit of ancient Indian traditions and culture accessible to modern younger generation of the world through research and development of cutting edge technologies

Team Details

Bharath N	rkmitddev@gmail.com	7904822572
V Sugandhan	rkmlabot@gmail.com	8939152777
Ragul Venkadesan	rkmitdbot@gmail.com	6382691891
Venkatesh D	venkat@zohocorp.com	9840718049

Team: SatyaBharat

Team Size: 3

Mentors: 1

Organization: Ramakrishna Math Media Lab

Type: Professionals

References

- 1) Nanda, P. K. K., Pretika. (2020). Every house a school, every parent a teacher as Covid-19 impacts education. *Live Mint*.
- 2) Singh, S., Roy, D., Sinha, K., Parveen, S., Sharma, G., & Joshi, G. (2020). Impact of COVID-19 and lockdown on mental health of children and adolescents: A narrative review with recommendations. *Psychiatry Research*, 293, 113429.
- 3) Mahamedhananda, Srinivas (2018). Vivekananda Way. *The Vedanta Kesari*.
- 4) Seema, P., & Ravi, P. (2020). *Indian Toys - Toys for Children*, from <http://www.dsourc.in/resource/indian-toys/social-toys>
- 5) Guha, M. S., & Murthy, M. K. (2020). Understanding the role of toys in child development: Special focus on traditional channapatna toys.
- 6) Hurd, N. M., & Zimmerman, M. A. (2011). Role Models. In R. J. R. Levesque (Ed.), *Encyclopedia of Adolescence* (pp. 2399-2404). New York, NY: Springer New York.
- 7) Singer, D. G. (1994). Play as healing. In J. H. Goldstein (Ed.), *Toys, Play, and Child Development* (pp. 147-165). Cambridge: Cambridge University Press.
- 8) Jiloha, R. C. (2010). Deprivation, discrimination, human rights violation, and mental health of the deprived. *Indian journal of psychiatry*, 52(3), 207-212.
- 9) OKONKWO, N. C. (2020). SMART TOYS: PREREQUISITE FOR DEVELOPMENT IN TODDLERS. *National Journal Of Educational Leadership*, 5(2).

References

- 10) Cagiltay, K., Kara, N., & Cigdem, C. (2014). Smart Toy Based Learning. In (pp. 703-711).
- 11) Wise, R. (2019, 2019-08-16). *How Do Positive Role Models Affect Our Youth and Communities?*, from <https://educationandbehavior.com/assess-the-value-of-positive-role-models-on-ones-behavior/>
- 12) Gozali-Lee, E. M. (1994). Older siblings as positive role models in the development of at-risk children.
- 13) Tang, Y.-Y., Jiang, C., & Tang, R. (2017). How Mind-Body Practice Works—Integration or Separation? *Frontiers in Psychology*, 8.
- 14) Johnson, E., González, I., Mondéjar, T., Cabañero-Gómez, L., Fontecha, J., & Hervás, R. (2020). An Affective and Cognitive Toy to Support Mood Disorders. *Informatics*, 7(4), 48.
- 15) Greenberger, E., Chen, C., & Beam, M. R. (1998). The Role of “Very Important” Nonparental Adults in Adolescent Development. *Journal of Youth and Adolescence*, 27(3), 321-343.
- 16) Chakraborty, P. (2009). *Swami Vivekananda's Childhood: Advaita Ashrama*.
- 17) Imani, M., & Montazer, G. A. (2019). A survey of emotion recognition methods with emphasis on E-Learning environments. *Journal of Network and Computer Applications*, 147.
- 18) Spence, S. H., Barrett, P. M., & Turner, C. M. (2003). Psychometric properties of the Spence Children's Anxiety Scale with young adolescents. *Journal of Anxiety Disorders*, 17(6), 605-625.
- 19) Serengil, S. I., & Ozpinar, A. (2020). LightFace: A Hybrid Deep Face Recognition Framework, 2020 *Innovations in Intelligent Systems and Applications Conference (ASYU)* (pp. 1-5).
- 20) Singh, S., Roy, D., Sinha, K., Parveen, S., Sharma, G., & Joshi, G. (2020). Impact of COVID-19 and lockdown on mental health of children and adolescents: A narrative review with recommendations. *Psychiatry Research*, 293, 113429.